

Marketing A/B Test Analysis

Ad Campaign vs. PSA (Public Service Announcement): Conversion Impact

Overview

This analysis evaluates whether showing users a paid advertisement, rather than a public service announcement (PSA), increases conversion rate. The dataset contains approximately 588,000 users randomly assigned to an **ad** group or a **psa** (control) group, with conversion outcomes and exposure metadata (ads seen, day, hour) recorded for each user.

Business Question

Does the ad campaign produce a statistically significant and practically meaningful increase in conversion rate compared to the PSA, and is that effect consistent across days of the week?

Method

- **Tools:** Python (pandas, SciPy, seaborn, matplotlib) for analysis; Power BI for the interactive dashboard.
- **Significance test:** Chi-square test of independence, comparing conversion outcome by test group.
- **Confound check:** Welch's t-test (unequal variance) comparing total ads seen between groups, to rule out exposure volume as an alternative explanation for any conversion difference.
- **Effect size:** Absolute lift (percentage-point difference) and relative lift (%), with a 95% confidence interval for the difference in proportions.
- **Segmentation:** Conversion rate and lift recalculated by day of week and hour of day to test consistency of the effect.
- **Multiple comparisons correction:** Because segmentation involves running many significance tests at once (7 for day of week), a Bonferroni correction was applied to control the overall false-positive rate. Without correction, running 7 tests at the standard 5% threshold carries roughly a 30% chance of at least one false positive purely by chance.

Key Results

Metric	Value
Ad conversion rate	2.55%
PSA conversion rate	1.79%
Absolute lift	0.77 percentage points
Relative lift	43.09%

95% CI for absolute lift	0.60% – 0.94%
Overall significance (chi-square)	$p < 0.00001$
Confound check (ads seen, t-test)	$p = 0.827$ (no significant difference)

The ad group converted at 2.55% versus 1.79% for the PSA group, an absolute lift of 0.77 percentage points and a relative lift of 43.09%. The chi-square test confirms this difference is highly unlikely to be due to chance ($p < 0.00001$), and the 95% confidence interval (0.60%–0.94%) indicates the true lift is very unlikely to be zero or negative. The confound check shows no significant difference in total ad exposure between groups ($p = 0.827$), which rules out ad volume as an alternative explanation for the result.

Segment-Level Findings: Day of Week

Conversion lift was recalculated for each day of the week and tested individually, then compared against a Bonferroni-corrected significance threshold ($\alpha = 0.05 / 7 = 0.00714$) to control for the increased false-positive risk of running multiple tests.

Day	Ad Conv.	PSA Conv.	Significant after correction?
Monday	3.32%	2.26%	Yes
Tuesday	3.04%	1.44%	Yes
Wednesday	2.54%	1.58%	Yes
Thursday	2.16%	2.02%	No
Friday	2.25%	1.63%	No
Saturday	2.13%	1.40%	No
Sunday	2.46%	2.06%	No

Only Monday, Tuesday, and Wednesday show a lift that remains statistically significant after correction. Thursday through Sunday show positive raw lift, but the difference on those days is not statistically distinguishable from random variation at this sample size. Reporting all seven days as "significant" without this correction would overstate the reliability of the day-level finding.

Recommendation

Adopt the ad campaign over the PSA: the overall lift is both statistically significant and practically meaningful (43% relative improvement in conversion). For day-targeted strategies, confidence should be limited to Monday–Wednesday, where the effect holds up under a corrected significance threshold; claims about weekend or Thursday/Friday performance should be treated as inconclusive pending a larger sample.

Limitations & Next Steps

- Hour-of-day segmentation (24 tests) carries an even higher multiple-comparisons risk than the day-level analysis and should be corrected and reported separately before any hour-specific claims are made.
- Segments with very small PSA sample sizes (e.g. specific hour/day combinations) may not meet the minimum expected cell count assumption for the chi-square test and should be flagged as lower-confidence if reported.
- This analysis measures conversion lift only; it does not assess cost per acquisition or return on ad spend, which would be needed for a full business case.